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لجنة الميكانيك - الإتجاه الإسلامي

Al-Balqa Applied University
Engineering Economy final Exam

Student Name.....

Time: 100 min.

Use the given data in the following Table to solve Q1 to Q3

Items	Alt.A ✓	Alt.B ✗	Alt.C ✓	Alt.D ✗
F.C	140,000	-	15,000	65,000
Extra cost @end of 3 rd yr	20,000	-	4,000	-
AOC	-	-	1,000	2,500
A.C for the first 5yrs	5,500	-	-	-
Cost every 6 months	-	500	-	-
Extra cost every 5 yrs	8,000	-	-	-
A.I for the first 5yrs with decreasing gradient (G=400)	-	-	3,000	-
S.V	-	-	7,000	5,000
Life(n) yrs	∞	7	5	4
i%	8%	20% per yr compounded quarterly	8%	MARR=10%

Q1: Using the capitalized cost criterion which is the optimal alternative A or C? (15 marks)

a- CC1(A) is

b- CC2(A) is

c- CC3(A) is

d- CC4(A) is

e- CC (A) is

f- CC1(C) is

g- CC2(C) is

h- CC3(C) is

i- CC4(C) is

j- CC5(C) is

k- CC(C) is

Q2: The future worth for alternative B is



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Q3: Is alternative D

1- The value of (i^*) is:.....

2- Is alternative D Acceptable or not ..

(10 marks)

Q4: If JD 2000 is deposited each semiannual for 5 years at an interest rate of 10 % yearly compounded semiannually , the future worth is.....

(10 marks)



Q5: The total amount due after 7 years for a loan of 30000 at 20% simple interest rate per year is.....

(5 marks)



Polytechnic



لجنة الميكانيك - الإتجاه الإسلامي

Al-Balqa Applied University
Faculty of Engineering Technology
Civil Engineering Department
Engineering Economy final Exam 2010/2011

Lecture time

Student Name..

Time: 2H.

(B)

Use the given data in the following Table to solve Q1 to Q3

Items	Alt.A	Alt.B	Alt.C	Alt.D	Alt.E
F.C	8,000	11,000	5,000	400	200
Extra cost @end of 3 rd yr	300	-	100	-	-
AOC	-	-	-	-	-
A.C for the first 5yrs	50	-	-	-	-
Cost every 6 months	-	500	-	-	-
A.I for the first 5yr	-	120	-	120	75
Extra cost every 5 yrs	70	-	100-	-	-
A.I for the first 5yrs with decreasing gradient (G=40)	-	-	300	-	-
S.V	-	1,000	700	150	100
Life(n) yrs	∞	7	5	8	4
i%	13%	12% per yr compounded Quarterly	13%	MARR=25%	MARR=25%

ANSWER ANY FIVE QUESTION ONLY.

Q1: Using the capitalized cost criterion which is the optimal alternative A or C?

- a- CC(A) is -8675,62 8466,865 (4 marks)
b- CC(C) is -11974,52 (4 marks)
c- The decision is... A (2 marks)

Q2: a-The future worth for alternative B is..... (5 marks)

b- The AW for alternative C is (5 marks)

Q3: Using the Rate of return (IRR) which is the optimal alternative D or E?

a- ROR(D) is 54.97.2 (4 marks)

b- ROR(E) is 5.34 (4 marks)

c- The decision is... D (2 marks)

Q4: a) If JD 200 is deposited each quarter for 6 years at an interest rate of 10 % yearly compounded continuously, the present worth is..... (5 marks)

b) The total amount due after 9 years for a loan of 9000 at 12% simple interest rate per year is..... (5 marks)



لجنة الميكانيك - الإتجاه الإسلامي

Use $i=12\%$

Use the given data in the following Table to solve Q5

Cash Item	F	G
Initial (first) cost	1800	2000
Annual O.C	650	18000
Annual Income or Users income	790	20000
S.V	1200	1100
Life (years)	6	3

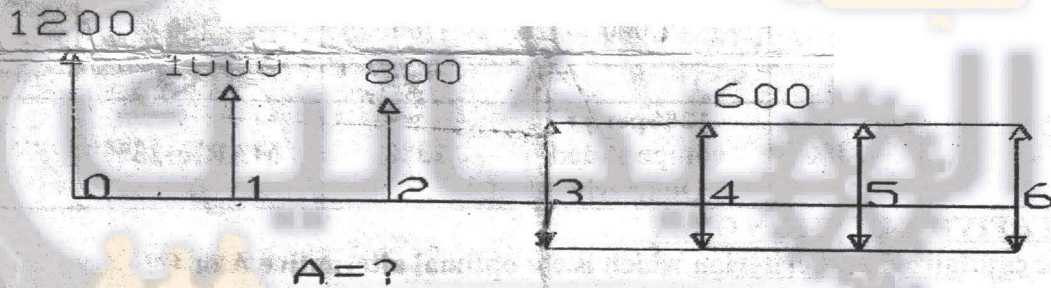
Q5: a). For Alt F, using the SOYD method
The depreciation at the end of year 5 is ~~1228.57~~ 57.14 (5 marks)

b). For Alt. G, using SL method:

The book values at years 3 and 7 are:

$BV_3 = \frac{2000 - 1100}{3} = 300$ $BV_7 = 1100$ (3 marks)

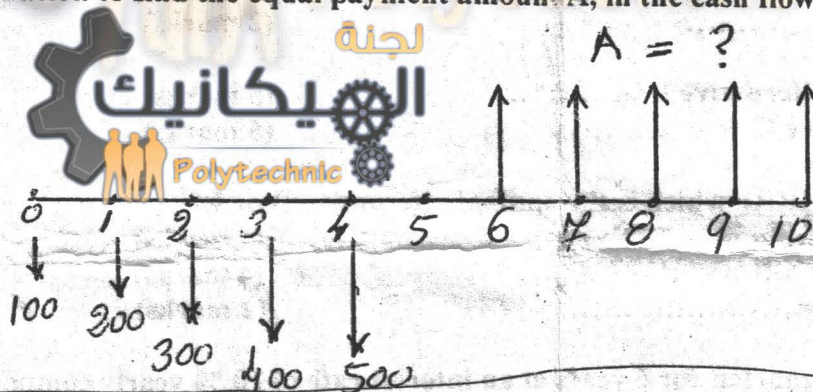
Q6: a). Find the Equivalent Values (A), (Use $i=8\%$ per year). (5 marks)



$A = ?$

$A = 1505.76$

b). the equation to find the equal payment amount A, in the cash flow diagram at $i=12\%$ (5 marks)



- a) $A(P/A, 12\%, 5) = 100(F/A, 12\%, 5) (F/P, 12\%, 1) + 100(P/G, 12\%, 5) (F/P, 12\%, 6)$
 b) $A(P/A, 12\%, 5) = 100(P/G, 12\%, 6) (F/P, 12\%, 7)$
 c) $A(F/A, 12\%, 5) = 100(F/A, 12\%, 5) (F/P, 12\%, 6) + 100(P/G, 12\%, 5) (F/P, 12\%, 11)$
 d) All of the above
 e) Non of the above

